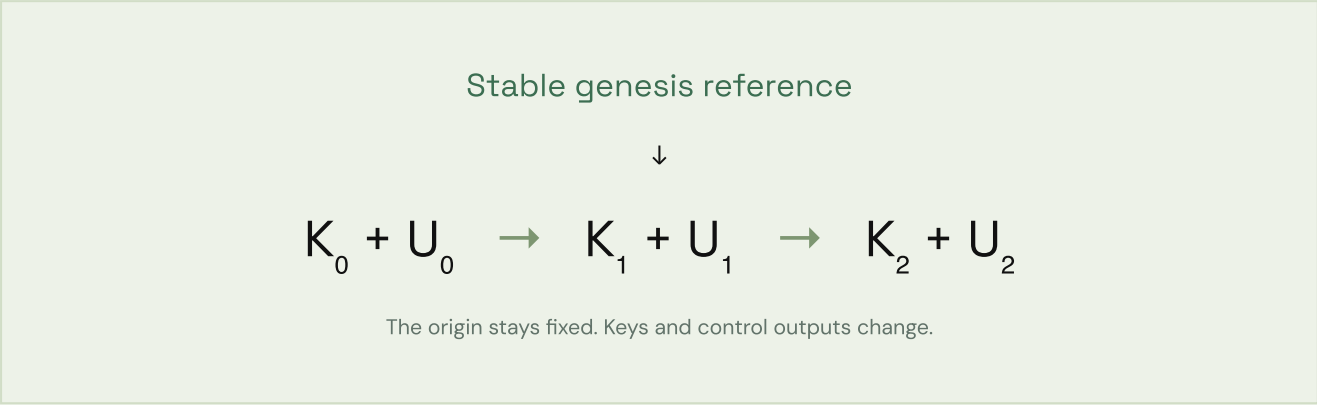
rooted.

A BITCOIN IDENTITY RESEARCH PROPOSAL

Identity that outlives its keys.

A Bitcoin-anchored model for key continuity, recovery, and revocation.



Original idea: Michael Tidwell

Version 0.1 · Research draft · October 19, 2026

Developed from a supplied brainstorming sketch. The protocol formulation and technical review are editorial proposals pending Michael's review. This is a design study, not a production protocol or an independent security audit.

`rooted-identity.vercel.app · github.com/sntslol/rooted-identity`

A proposal, ready to question. This is a research draft, not a deployed identity protocol. The original idea is Michael Tidwell's; the protocol details and technical review are editorial proposals, pending his review.

01 Abstract

A public key proves that a message was signed by someone with a particular secret. It does not tell an application what to do when that secret is lost, replaced, or stolen.

Rooted proposes a stable identity reference anchored to a Bitcoin transaction output. A compact OP_RETURN record binds an application signing key to a Taproot control output. Spending that output advances the identity to a new key and control output, explicitly revokes it, or ends its recognized continuity. The original reference stays the same.

Bitcoin supplies transaction ordering and spend authorization. Rooted clients supply the rules that interpret those transactions as identity changes. Taproot can express owner and guardian spending conditions; it does not itself understand identity, reputation, or rotation. The design uses existing Bitcoin primitives and proposes no consensus change.

The defensible claim: a verifiable history of authorized key succession. That history is not proof of a human being, a unique person, good behavior, or uninterrupted control by the same individual.

02 The problem: a key is not a lifetime

Imagine a player who has spent years building a reputation in a game. Their account is a public key. They want to replace a laptop, recover from losing a device, or retire a leaked signing key without starting from zero.

A server can maintain an account database, but that server becomes an authority for the account's continuity. A bare public key removes that authority while leaving recovery and succession to each application.

The proposal separates three things:

- **Identity reference:** the permanent pointer to the identity's genesis output.
- **Application key:** the current key for messages, game actions, or authentication.
- **Control policy:** the Bitcoin spending conditions that authorize the next identity state.

The application key should normally be separate from Bitcoin control keys. Losing a frequently used game or Nostr key then need not give an attacker the ability to rotate the identity. A control policy can remain offline until a change is needed.

A domain name, profile, avatar, or payment address can be associated with this reference. None of those associations establishes a legal identity or a unique person.

03 The model: one identity, one live output

The baseline has one identity per control UTXO: an unspent transaction output. Its stable reference is a tuple of **network genesis-block hash, genesis transaction ID, and control-output index**. Network binding prevents the same-looking reference from being interpreted on the wrong chain. The displayed shorthand in this site is illustrative; no textual identifier standard is assigned.

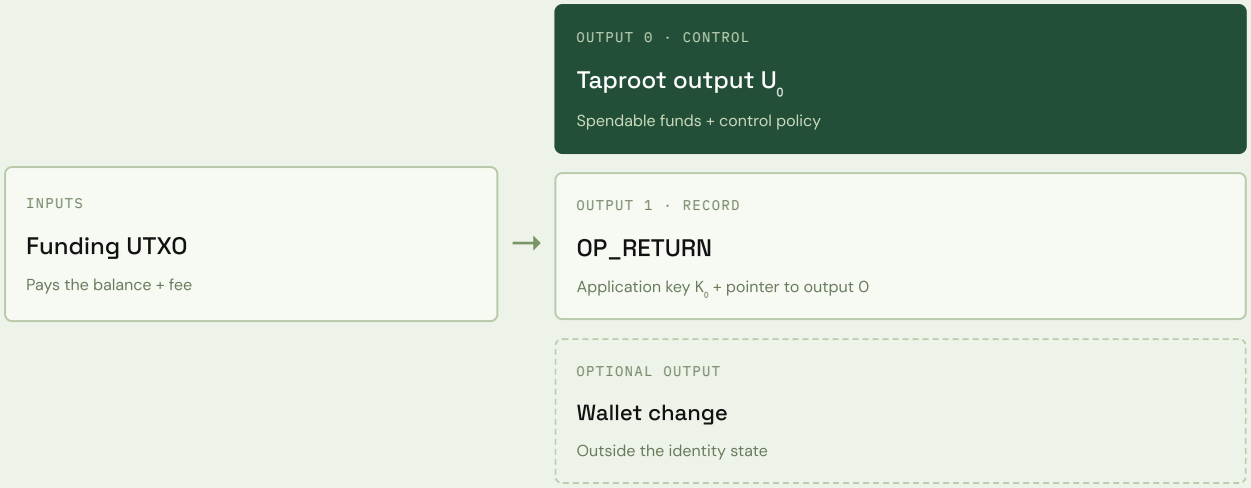
A creation transaction contains a zero-value OP_RETURN record and a funded Pay-to-Taproot control output. The record identifies that output and the initial application key. Change outputs and fee inputs are ordinary Bitcoin accounting, not identity state.

The Taproot output commits to an internal public key and a tree of spending conditions. A wallet can construct it with a verifiably derived NUMS internal point whose discrete logarithm is unknown, making the key path computationally unavailable under the usual cryptographic assumptions. It should retain evidence of that construction. The address alone cannot prove that no one knows a key-path secret. See BIP 341 (https://bips.dev/341/).

The application key appears in the record; it need not be one of the spending keys. A Taproot address is an encoding of an output key, not a store of readable scripts. An unspent output hides its script tree. A script-path spend reveals the executed script, its inputs, and a control block, while unused branches stay hidden.

> **Bitcoin checks who may spend. Rooted checks what that spend means.**

FIG. 02 Inside a creation transaction



The genesis reference identifies output 0. The record points to it; the application key and the control keys can be different.

04 The lifecycle: create, rotate, recover, revoke

Create. Confirm a transaction binding key K_0 to control output U_0. Its outpoint becomes the permanent reference. Creation alone claims no history before that transaction and proves no ownership of a real-world name.

Rotate. Spend the current control output. The same transaction carries a valid rotation record, the new application key K_1, and exactly one designated successor control output U_1. The resolver follows the spent outpoint to that successor. The identity reference remains unchanged.

Recover. Use an alternative authorized Taproot script, such as two of three guardians, to make the same rotation. Recovery is a spending-policy choice, not a separate on-chain operation. A guardian quorum has control power; it can also abuse that power.

Revoke. Spend the current output with an explicit revocation record and no successor. In this proposal, revocation is terminal on the selected chain. A new genesis creates a different identity; a link to it is an assertion, not restoration of the old one.

Unrecognized spend. If complete, validated transaction data shows that the current output was spent without a valid continuation or revocation, the v1 identity becomes terminated. Do not keep treating its old application key as current. If the data is unavailable, the resolver reports an unresolved state instead of inventing a transition.

A reorganization can undo any of these outcomes until the relevant transactions are sufficiently buried. “Terminal” refers to the history of the currently selected chain, not absolute finality.

05 A concrete transaction record

To make the idea reviewable, this paper proposes a deliberately narrow, fixed-width v1 record. This is a candidate encoding, not an implemented network protocol. One transaction carries exactly one Rooted record and manages exactly one identity. Batching is deferred.

All integer fields below use unsigned little-endian encoding. The payload is exactly **78 bytes**. It is pushed as a single item in an `OP_RETURN OP_PUSHDATA1 0x4e <payload>` script, which is 81 bytes long. With an 8-byte zero value and a 1-byte script length, that output occupies 90 non-witness bytes.

Offset	Field	Bytes	Meaning
0	Magic	4	ASCII `ROOT`; provisional namespace
4	Version	1	`0x01`
5	Operation	1	`0x00` create, `0x01` rotate, `0x02` revoke
6	Control input	2	Input index spending the current output; `0xffff` at creation
8	Successor output	2	Index of the new P2TR output; `0xffff` on revocation
10	State sequence	4	Zero at creation; previous sequence + 1 thereafter
14	Application key	32	Valid BIP 340 x-only public key; all zeroes on revocation
46	Document hash	32	SHA-256 of exact associated-document bytes; all zeroes if absent

The document hash can bind a profile or a policy manifest. Availability is separate from integrity: Bitcoin does not host the document. Version 1 commits to exact bytes, avoiding an implied JSON canonicalization standard. The document must carry its own schema and context; application signatures must separately bind their protocol and network.

An active successor must have a native P2TR scriptPubKey (`OP_1` followed by a 32-byte output key that lifts to a secp256k1 point). The index must be in range. Sequence overflow ends the v1 lineage; it must never wrap to zero.

For creation, require operation 0, sequence 0, control input `0xffff`, a valid application key, and a valid successor. For rotation, require operation 1, the exact current outpoint at the referenced input, sequence + 1, a valid new key, and a valid successor. For revocation, require operation 2, that same current input, sequence + 1, successor `0xffff`, and zero key and document fields. Reject extra or duplicate Rooted records and noncanonical framing.

A complete transition also requires a Bitcoin-valid script-path spend of the control output. A key-path spend is an unrecognized spend under this v1 profile. Wallets should sign control inputs with `SIGHASH_DEFAULT` or `SIGHASH_ALL` so their signatures commit to the complete output vector, including the record and successor. Weak sighash modes need separate security analysis and are outside the proposed wallet profile.

These are **client rules**, not covenants. The scripts shown here do not force a spender to create a successor, preserve funds, or publish this record. A spend can be valid Bitcoin and still terminate a Rooted identity. Never describe the output as consensus-enforcing the full identity state machine.

The 78-byte payload is intentionally small enough for the historical 83-byte data-carrier script policy. That policy was never an 80-byte consensus cap: Bitcoin Core 30.0 changed its defaults to permit larger and multiple OP_RETURN outputs. Relay and mining settings vary. See the Core 30.0 release notes (https://bitcoincore.org/en/releases/30.0/).

06 What Taproot actually contributes

Taproot offers a compact commitment to alternative ways of authorizing a spend. An illustrative control policy might offer an owner branch and a two-of-three guardian branch. Bitcoin accepts a spend satisfying either branch. The identity client then interprets its record.

An illustrative guardian leaf is:

```
<G1> OP_CHECKSIG
<G2> OP_CHECKSIGADD
<G3> OP_CHECKSIGADD
OP_2 OP_NUMEQUAL
```

Here G1–G3 are valid 32-byte x-only secp256k1 public keys. Witness signature positions are supplied in reverse key order, with empty elements for nonsigners. This is explanatory script, not a production descriptor or wallet implementation. Tapscript disables legacy CHECKMULTISIG and supplies CHECKSIGADD; see BIP 342 (https://bips.dev/342/).

A timelocked guardian branch can add `OP_CHECKSEQUENCEVERIFY` or `OP_CHECKLOCKTIMEVERIFY`. Relative delays require the transaction-version and sequence conditions of BIP 68 (https://bips.dev/68/) and BIP 112 (https://bips.dev/112/); absolute constraints follow BIP 65 (https://bips.dev/65/). These conditions make a spend eligible. They do not schedule a transaction or notify its owner.

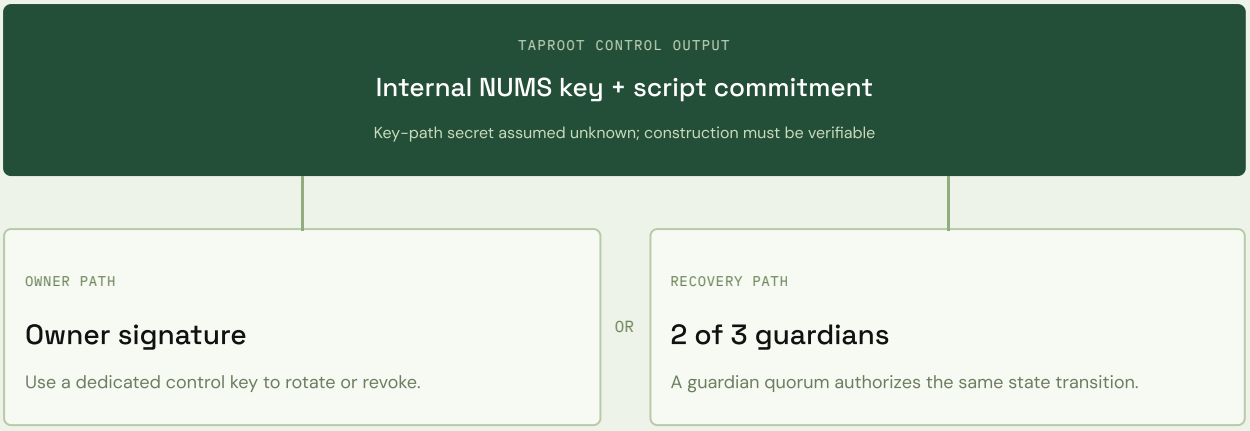
A simple delayed guardian branch does not provide a veto window after a recovery request: its delay runs from the control output's confirmation. Once eligible, it can race an owner spend. A real challenge-and-cancel recovery flow would need additional transaction stages and a separately analyzed policy.

Large threshold policies are possible in principle, subject to script, stack, signature-budget, weight, and relay limits. "Arbitrary quorum" is not a blanket assurance. A two-of-three policy is a better first prototype than an untested 20-of-30 design.

Unused leaves remain private, but a revealed CHECKSIGADD branch exposes the public keys in that leaf and the signature slots that succeeded. A specialized threshold signature can hide individual participation behind one aggregate key, with additional setup and nonce-safety requirements. MuSig2 is an n-of-n

construction, not a drop-in arbitrary threshold protocol; see BIP 327 (https://bips.dev/327/). Shamir secret sharing alone does not provide threshold signing: reconstruction brings a full secret together.

FIG. 03 Alternative ways to authorize a spend



Only the executed leaf and its proof are revealed. The two branches authorize spending; neither forces the spender to preserve a successor or its recovery policy.

07 Verification: follow the spend, verify the history

A resolver starts with the network-bound genesis outpoint, validates its creation record, and follows each confirmed spend of the current control output. At each step it checks the Bitcoin transaction, relevant script-path witness, canonical record, input binding, sequence, and successor. Ordinary inputs or change outputs do not advance the identity.

```
resolve(network, genesis, validated_chain_tip):
  state = validate_creation(genesis)
  while current_control_output_has_a_confirmed_spend(state):
    tx = obtain_complete_validated_spend(state.outpoint)
    if tx data is unavailable: return UNRESOLVED
    if recognized_revocation(tx, state): return REVOKED
    if not recognized_rotation(tx, state): return TERMINATED
    state = successor_state(tx, state)
  return ACTIVE(state, validated_chain_tip)
```

“Validated” does substantial work here. The recommended reference resolver uses a fully validating Bitcoin node plus an index of spends and Rooted records. A cached index is reconstructible, but still necessary infrastructure for efficient discovery. A pruned node may need external historical blocks and witnesses to rebuild it. Bitcoin Core’s transaction index alone is not a general outpoint-spend index.

A light client can check inclusion proofs, but a conventional txid Merkle proof does not authenticate witness bytes. Witness authentication needs the SegWit commitment through the coinbase and the associated proof material, or a stronger trusted/full-validation source. BIP 141 (https://bips.dev/141/) defines that commitment. Such proofs also do not establish that no later spend was omitted. Headers alone cannot establish the latest identity state.

A scan beginning at an advertised launch height discovers only identities created at or after that height. It is an explicit protocol boundary or a checkpoint assumption, not a way to reconstruct arbitrary older histories from headers.

Each resolver response should include the network, validated tip hash and height, identity genesis, current outpoint, current application key, sequence, observed state, confirmation count, and data-source assumptions. Clients should display pending changes separately and retain reversible state for reorgs. This draft suggests six confirmations as a configurable example for sensitive actions, not a finality guarantee or a universal security threshold.

08 Compromise, recovery, and trust

Application-key theft. An attacker can forge new application messages until clients recognize a confirmed rotation. If control keys are separate, stealing the application key alone cannot spend the identity output. Rotation limits future acceptance; it cannot erase previously signed content or reliably date when a secret was stolen. An attacker can backdate an ordinary signed timestamp. Applications need authenticated historical key state and independent time evidence for historical claims.

Control-key theft. An attacker who satisfies a control branch can rotate to their own keys or terminate the lineage. If an independent recovery branch still applies to the current output, the owner may race them. After a malicious rotation confirms to an attacker-only successor, the original guardians have no inherent authority over that new output. Basic Taproot scripts do not carry a mandatory recovery branch forward.

Unproven key claims. Anyone can publish another person's application public key in a creation record. The anchor alone does not prove possession of that key or authorize taking over an existing account. Applications should authenticate onboarding, pin the intended genesis reference, and require a domain-separated challenge signature from the application key. Never choose a user's identity by taking the first on-chain record that mentions their key.

Guardian collusion. Recovery authority is custody of a power, even when the guardians do not hold a conventional wallet balance. Choose quorums and backups accordingly. A visible guardian key shows an authorization relationship, not an endorsement or a social trust score.

Bounty output. Funding the output with 50,000 sats creates something an attacker may take; it does not guarantee disclosure, detect application-key theft, or distinguish theft from an authorized spend. It can increase attack incentives. Keep bounties outside the baseline security argument.

Loss and outages. Lose all viable signing material or all necessary script/control-block backups and control may become inaccessible. An unavailable indexer should produce an unresolved result. It must not cause a client to silently accept an old key as current.

Privacy and Sybil resistance. Public succession makes keys linkable. Explicit guardian scripts may reveal relationships. Fees increase creation cost but do not establish one person per identity. Avoid placing personal information or reusable payment addresses in permanent metadata.

Backups. Retain the full policy descriptor or equivalent tree data, derivation information, all needed scripts and control blocks, relevant key material or signer access, and state-discovery instructions. BIP 386 (https://bips.dev/386/) describes Taproot output descriptors. Encrypt sensitive backups and test recovery before funding. “Save the address” is not a recovery plan.

09 Useful first: games, Nostr, and agents

A small application is the right proving ground. A game can attach inventory history and reputation to the stable reference, accept actions signed by the current application key, and support an explicit recovery process. It should resolve state at login and before high-value actions. Users can create and recover test identities without putting mainnet funds at risk.

For Nostr, this is a proposed optional succession layer. Existing clients use signed events and public keys under NIP-01 (https://github.com/nostr-protocol/nips/blob/master/01.md). They will not automatically transfer followers or interpret a Bitcoin rotation. A client extension and explicit follow policy would be required. Do not equate “one key” with “one-time identity.”

NIP-05 (https://github.com/nostr-protocol/nips/blob/master/05.md) maps an internet identifier to a Nostr key through a domain’s well-known endpoint. A Bitcoin anchor could complement discovery, but is not itself NIP-05, a human-readable name, or proof of a domain. An endpoint outage does not invalidate Nostr signatures or erase the underlying key identity.

An AI agent could use the same succession reference across software upgrades, but continuity proves authorized control, not that the same model, operator, behavior, or intentions persist. Payment endpoints and other-chain identifiers should be typed, signed bindings with explicit scope and expiry. A mapping to Liquid does not import Bitcoin’s security assumptions into Liquid or create cross-chain finality.

This sketch has substantial prior context. DID Core (https://www.w3.org/TR/did/) separates identifiers, controllers, and verification methods. Sidetree (https://identity.foundation/sidetree/spec/) and did:btcr2 (https://github.com/dcdpr/did-btcr2) explore ledger-anchored identity and updates. This proposal should be compared with that work before asserting a distinctive contribution. Taproot’s arrival does not imply earlier DID systems lacked rotation or recovery.

10 Costs, limits, and the next experiment

Every creation or update consumes blockspace. The live control output also occupies the UTXO set and holds spendable bitcoin. An OP_RETURN output is provably unspendable; it does not become a live UTXO. The control balance must satisfy the relevant dust and relay rules and a wallet needs a practical fee-funding strategy.

The record output alone occupies 90 vbytes. A realistic complete update also includes inputs, a Taproot witness, a successor, and possibly change. This site's fee calculator uses an **illustrative 250-vbyte transaction**, not a measured benchmark or live fee quote. At 10 sat/vB that would cost 2,500 sats, excluding the retained control balance. Actual weight depends on the selected leaf, Merkle depth, signers, funding, and outputs.

Start with a regtest wallet and resolver for one identity per transaction, one owner branch, and one two-of-three guardian branch. Test the complete lifecycle, malformed records, lost backups, competing spends, fee replacement, missing history, and reorganizations. Then integrate a game login that refuses stale or unresolved key state. Mainnet deployment of the protocol is not proposed at this stage.

Before interoperability, freeze the namespace and wire format; specify the full accepted policy and sighash profile; publish Bitcoin transaction, witness, and resolver test vectors; independently review scripts; and measure real fees and recovery usability. Batching, cross-chain continuity, threshold privacy, and formal DID-method integration remain separate research tracks.

The website includes a tested educational state-machine model and record codec. It does **not** contain a Bitcoin wallet, transaction signer, script interpreter, node-backed resolver, or an independent security audit. Its technical review is a source-based design review. The viable next step is an executable regtest prototype, not a claim of production readiness.

CHECK THE ASSUMPTIONS

What holds up. What needs care.

A source-based review of the original sketch.

Expand any claim to inspect the reasoning.

Supported

Anchor a key to a Bitcoin output

A transaction can bind a key and an output index in OP_RETURN. The proposed identity meaning is enforced by clients.

BIP 341 ↗ Bitcoin Core 30.0 ↗

Corrected

Taproot enforces rotation and revocation

Taproot validates spending conditions. The proposed record and resolver define the action and successor. A Bitcoin-valid spend can end continuity.

BIP 341 ↗ BIP 342 ↗

Qualified

A NUMS key forces a script reveal

A verifiably derived point with unknown discrete logarithm makes key-path use computationally unavailable. An address alone does not prove its construction.

BIP 341 ↗

Corrected

Timelocks enable automatic rotation

Timelocks constrain eligibility. Someone must still construct, sign, fund, and broadcast a transaction.

BIPs 65 / 68 / 112 ↗

Corrected

A delayed recovery branch provides a veto period

A simple relative lock runs from output confirmation, not from a later recovery request. Challenge-and-cancel recovery needs additional stages.

BIPs 65 / 68 / 112 ↗

Corrected

Guardian recovery is a web of trust

Guardian keys have control authority. Social endorsement and reputation require separate signed assertions and application rules.

BIP 342 ↗ DID Core ↗

Qualified

Recovery does not reveal which guardian signed

A revealed CHECKSIGADD leaf exposes its keys and signing slots. Hiding participants requires a separately analyzed aggregate or threshold construction.

BIP 342 ↗ BIP 327 ↗ RFC 9591 ↗

Corrected

Shamir shares are threshold signing

Secret sharing alone reconstructs a secret. It does not supply distributed signature generation or its nonce-safety guarantees.

RFC 9591 ↗

Corrected

An on-chain anchor is NIP-05

NIP-05 is a domain-to-key mapping. An anchor can complement discovery; existing Nostr clients do not inherit succession support.

NIP-05 ↗ NIP-01 ↗

Corrected

Headers are enough; no indexing is needed

Resolvers need spends, records, and relevant witnesses. Inclusion does not prove latest state; witness authentication needs more than a txid Merkle branch.

BIP 141 ↗

Qualified

Any public/private key can control the output

Taproot signature checks use BIP 340 secp256k1 keys. Version 1 application keys use the same format; other key types need an extension.

BIP 340 ↗ BIP 342 ↗

Corrected

A bounty proves a key was compromised

A spend proves the spending conditions were satisfied, not that theft occurred. An attacker may keep a stolen key secret or use a valid rotation.

BIP 341 ↗

Corrected

Recovery survives any malicious rotation

The successor policy can omit the old guardians. The baseline scripts do not enforce policy inheritance across outputs.

BIP 341 ↗ BIP 342 ↗

Corrected

80 bytes is an OP_RETURN consensus limit

That number described historical relay policy. Core 30.0 changed its defaults; deployment still depends on actual relay and mining policies.

Bitcoin Core 30.0 ↗

Corrected

Continuity proves the same person

It proves authorized succession under a policy. Keys and control can be transferred; multiple identities can belong to one operator.

DID Core ↗

Open

The design is ready for mainnet

A candidate record and educational model are provided. Regtest transactions, reviewed scripts, a real resolver, interoperability vectors, and independent review remain.

See the prototype roadmap →

Reviewed against primary sources on October 19, 2026. This is a design review, not an independent security audit or validation of deployed Bitcoin software.

FOLLOW THE EVIDENCE

Primary sources.

The specifications behind the claims.
Read them. Challenge our interpretation.

[01] **BIP 341**
Taproot spending rules, NUMS internal keys, script-path disclosure

[02] **BIP 342**
Tapscript, CHECKSIGADD, signature and execution limits

[03] **BIP 340**
Schnorr signatures, x-only keys, domain separation

[04] **BIP 141**
SegWit transaction IDs and witness commitments

[05] **BIPs 65 / 68 / 112**
Absolute and relative spending constraints

[06] **Bitcoin Core 30.0**
OP_RETURN relay-policy changes; policy versus consensus

[07] **NIP-01**
Nostr public keys and signed events

[08] **NIP-05**
Internet identifier mapping and client follow behavior

[09] **BIP 327**
MuSig2 n-of-n signing and protocol requirements

[10] **BIP 386**
Taproot output script descriptors

[11] **DID Core**
Identifiers, controllers, and verification methods

[12] **Sidetree**
Prior work on ledger-anchored identity and updates

[13] **did:btcr2**
Prior work on Bitcoin-anchored decentralized identifiers

[14] **RFC 9591**
Threshold signing, distinct from reconstructing a shared secret